

# VLSI Design and Testing

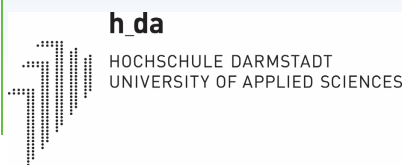
## Lecture WS 22/23

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**Faculty of Electrical Engineering  
and Information Technology**



VLSI Design and Testing

# VLSI Design and Testing

## OUTLINE

### **A: Combinational Circuit Design**

- CMOS-Inverter
- Basic gates in CMOS

### **B: Memory Circuit Design**

- Memory Architecture
- SRAM/DRAM/Flash

### **C: Design Methodology**

- Full custom design
- Semi custom design
- SoC/MPSoC design

### **D: Design Verification**

- ASIC-Prototyping
- FPGA-in-the-loop simulation

# VLSI Design and Testing

## **E: IC Fabrication**

Intro to CMOS Processing  
Yield, Manufacturing Costs

## **F: IC Testing**

Manufacturing Test Principles  
Design for Testability (DFT)

# VLSI Design and Testing

Textbook [1]:

N. Weste, D. Harris:  
**CMOS VLSI Design**  
A Circuits and Systems Perspective  
Fourth Edition (US), 2010, Addison-Wesley  
**Integrated Circuit Design**  
Fourth Edition (Global Edition) 2011, Pearson Education

for some specific topics please refer to [2]:

J. Rabaey, A. Chandrakasan, B. Nikolic:  
**Digital Integrated Circuits**  
A Design Perspective  
Second Edition, 2003, Prentice Hall

# VLSI Design and Testing

Textbook [3] for testing of VLSI circuits:

M. L. Brushnell, V. D. Agrawal:  
Essentials of Electronic Testing for Digital, Memory &  
Mixed-Signal VLSI Circuits  
Kluwer Academic Publishers, 2000

Textbook [4] for design of digital signal processing systems:

Keshab K. Parhi:  
VLSI Digital Signal Processing Systems: Design and  
Implementation , John Wiley & Sons, 1999.

## Microelectronics

Is the art, science and technology of designing and fabricating integrated circuits!

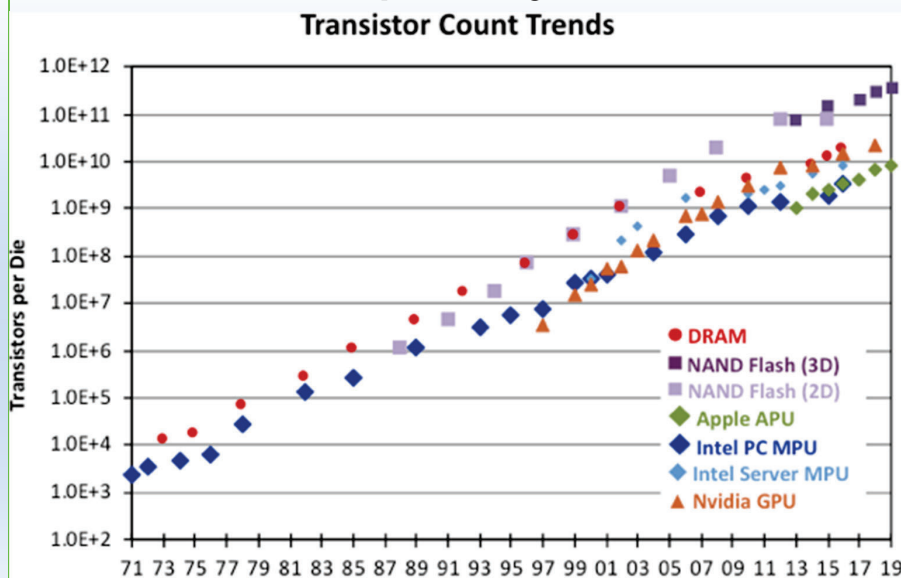
Areas of Microelectronics are:

- VLSI Design
- VLSI CAD Tools
- Modeling and Simulation
- Technology & Fabrication
- Testing

## Some keywords

- **Very-large-scale-integration** (VLSI) is defined as a technology that allows the construction and interconnection of large number of transistors on a single integrated circuit.
- **Integrated circuit** is a collection of gates fabricated on a single silicon chip.
- **Wafer** is a thin slice of semiconductor material on which semiconductor devices are made. Also called a slice or substrate.
- **Die** is an individual chip out of several identical chips that are produced after dicing up a wafer.

## Complexity : Moore's Law



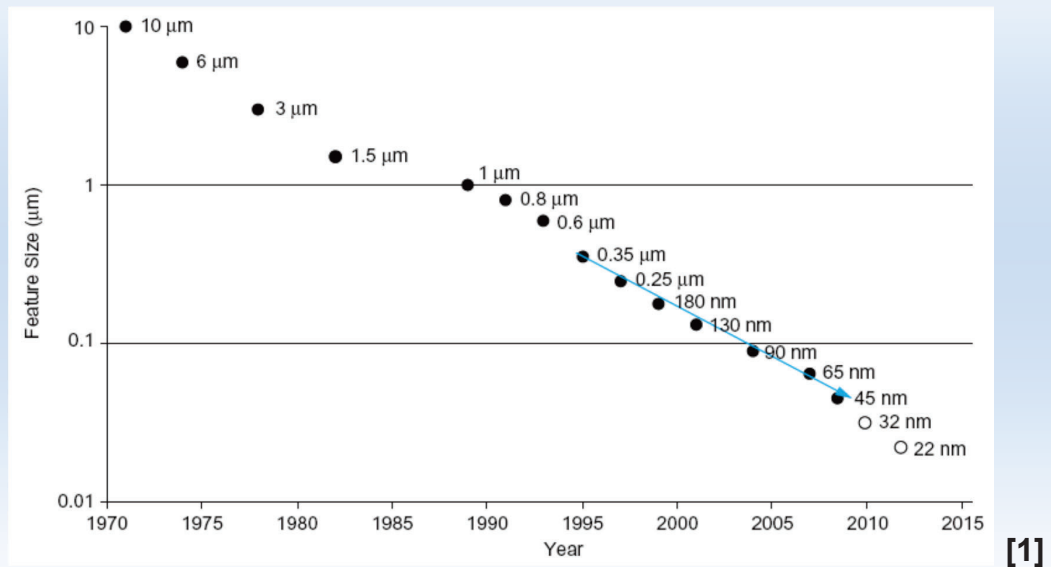
Sources: Intel, SIA, Wikichip, IC Insights

[IC Insights' 2020 edition]

Moore's Law was predicted by co-founder of Intel, Gordon Moore, in 1965 and slightly corrected in 1975.

# Feature Size

- Minimum feature size shrinking 30% every 2-3 years



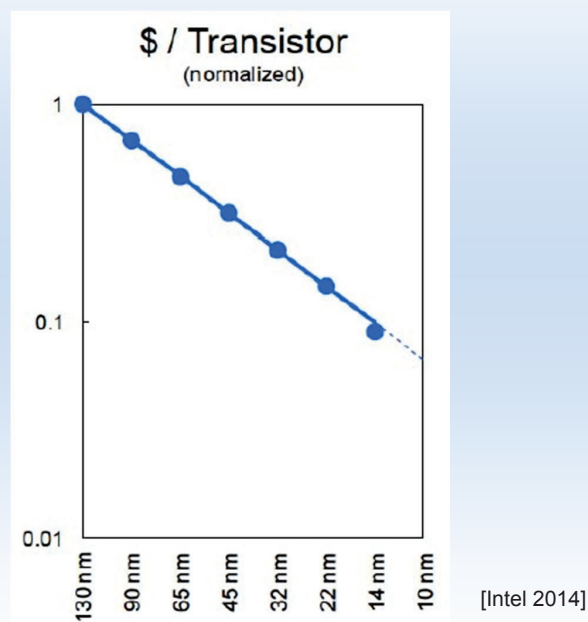
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# Cost per Transistor

- Driving force of feature size reduction



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## ❑ Metal Oxide Semiconductor Field Effect Transistors

- nMOS and pMOS MOSFETS
- Voltage applied to insulated gate controls current between source and drain
- Low power allows very high integration

## Moore's Law

### ❑ 1965: Gordon Moore

- Transistor counts have doubled approx. every 2 years and continues to do so!

### ❑ 1958: First integrated circuit

- Flip-flop using two transistors
- Built by Jack Kilby at Texas Instruments

### ❑ 2020

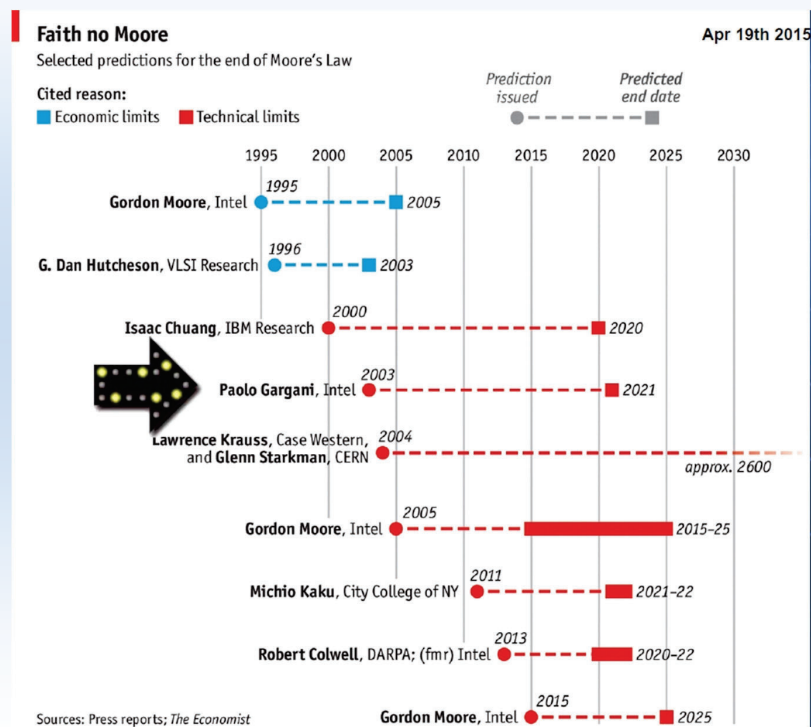
- Apple M1 SoC (16 billion transistors, 5nm CMOS process, 119mm<sup>2</sup> die size)
- Samsung 16 Gbit DRAM (> 16 billion transistors)

❑ Transistor counts in Intel's PC microprocessors grew approximately 40% per year through 2010, but the rate dropped to half that in the years following (power issue).

### ❑ Driven by miniaturization of transistors

- Smaller is cheaper, faster, lower in power!
- Revolutionary effects on society

# End of Moore's Law?

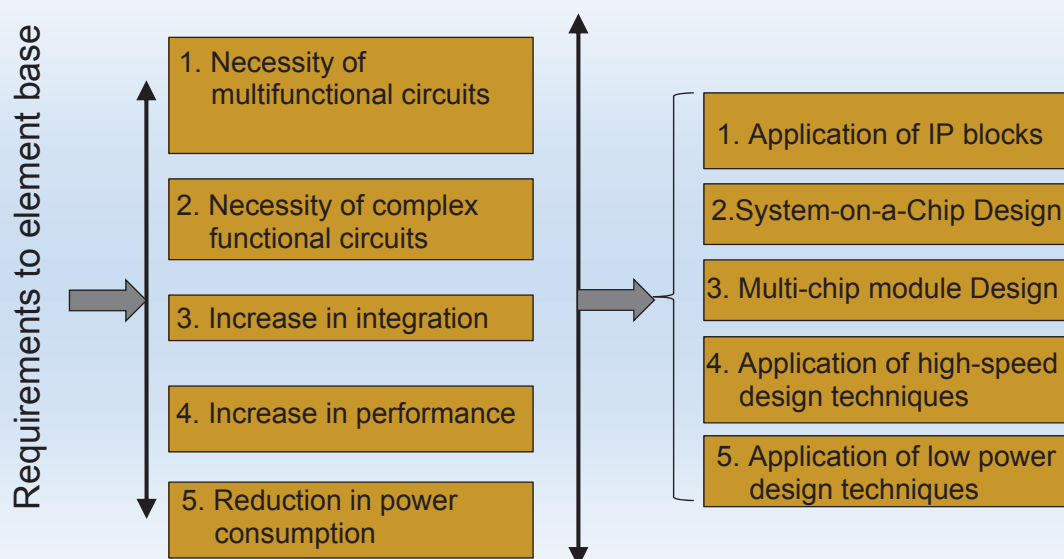


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## VLSI Development Trends (1)



[Synopsys]

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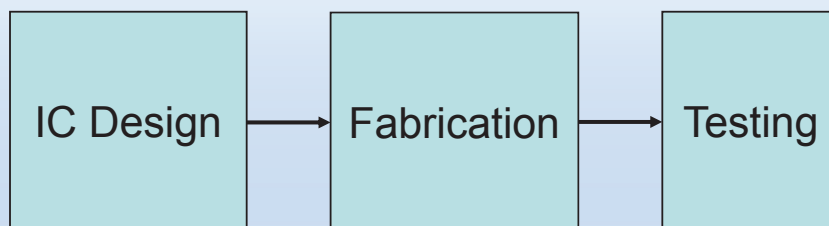
# VLSI Development Trends (2)

Improvement Trends for VLSI Enabled by Feature Size Scaling

| Trend             | Example                           |
|-------------------|-----------------------------------|
| Integration Level | Components/chip, Moore's Law      |
| Cost              | Cost per function                 |
| Speed             | Microprocessor throughput         |
| Power             | Laptop or cell phone battery life |
| Compactness       | Small and light-weight products   |
| Functionality     | Nonvolatile memory, imager        |

[Synopsys]

# VLSI Creation Cycle





# VLSI Design Problems (1)

The most important characteristics

- Area
- Speed
- Power dissipation
- Design time
- Testability

All these characteristics can be combined into a single cost function, the VLSI cost function.

It is impossible to try to design a VLSI circuit at one shot while at the same time optimizing the cost function. The complexity is simply too high.

Two main concepts that are helpful to deal with this complexity are [hierarchy](#) and [abstraction](#).

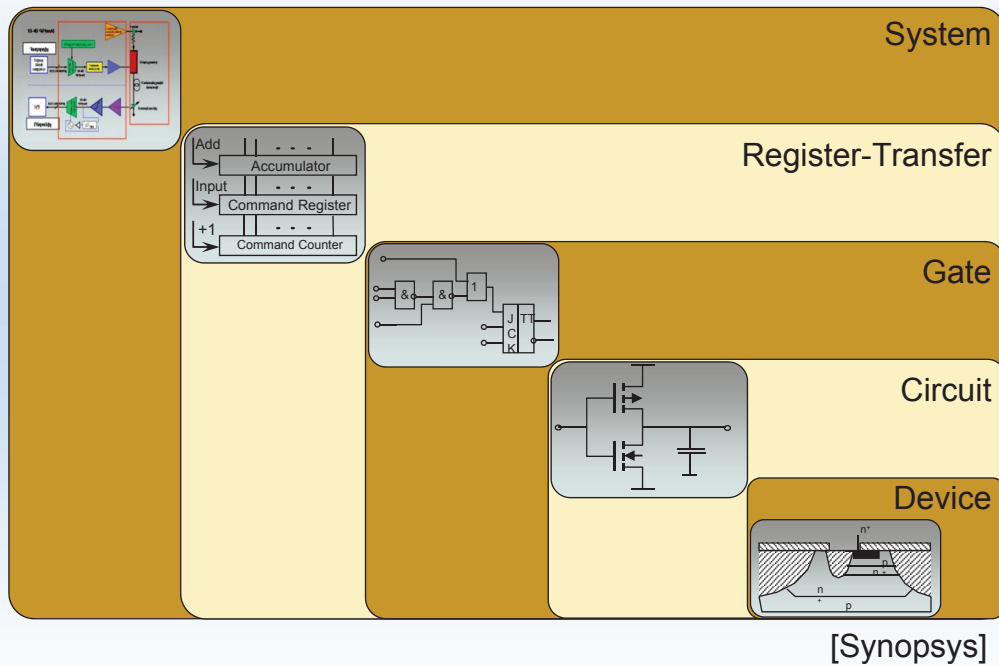
# VLSI Design Problems (2)

## Hierarchy and Abstraction

- Hierarchy shows the structure of a design at different levels of description.
- Abstraction hides the lower level details.

The use of abstraction makes it possible to reason about a limited number of interacting parts at each level in the hierarchy. Each part is itself composed of interacting subparts at a lower level of abstraction. This decomposition continues until the basic building blocks (e.g. transistors) of a VLSI circuit are reached.

# VLSI Design Abstraction Level (1)

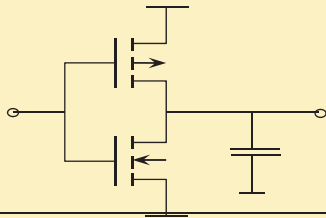
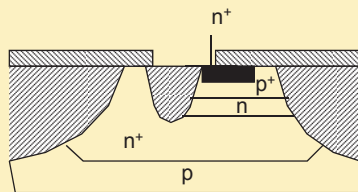


# VLSI Design Abstraction Level (2)

| N | Level                   | Modeling Object                           | Example of Modeling Object | Mathematical Apparatus      |
|---|-------------------------|-------------------------------------------|----------------------------|-----------------------------|
| 1 | System                  | Structural Circuit                        |                            | Theory of Massive Operation |
| 2 | Register-Transfer Level | Functional Circuits at the Register-level |                            | Boolean Algebra             |
|   | Logic                   | Circuit at the gate and flip flop-level   |                            |                             |

[Synopsys]

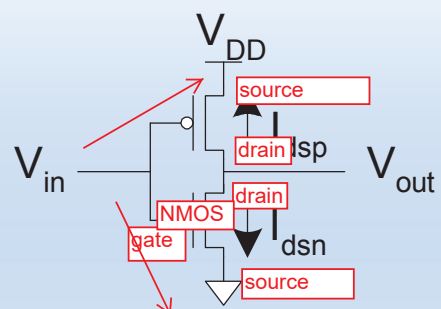
## VLSI Design Abstraction Level (3)

| N | Level     | Modeling Object    | Example of Modeling Object Level                                                   | Mathematical Apparatus                                                 |
|---|-----------|--------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| 3 | Circuit   | Electrical Circuit |  | System of differential equations                                       |
| 4 | Component | IC Components      |  | System of differential equations involving partial arbitrary functions |

[Synopsys]

## CMOS-Inverter

- When  $V_{in} = 0 \rightarrow V_{out} = V_{DD}$
- When  $V_{in} = V_{DD} \rightarrow V_{out} = 0$
- In between,  $V_{out}$  depends on transistor size and current
- It must hold:  $I_{dsn} = |I_{dsp}|$
- We could solve equations
- But graphical solution gives more insight

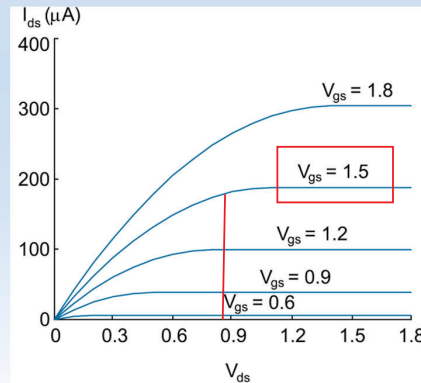


# CMOS-Inverter:I/O characteristic

❑ Current depends on region of transistor behavior

❑ For what  $V_{in}$  and  $V_{out}$  are nMOS and pMOS in

- Cutoff?
- Linear?
- Saturation?

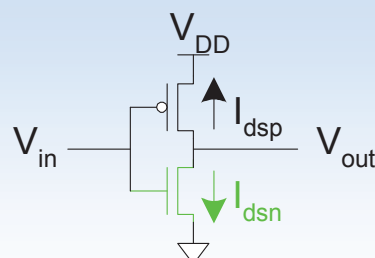


**FIG 2.7** I-V characteristics of ideal nMOS transistor

[1]

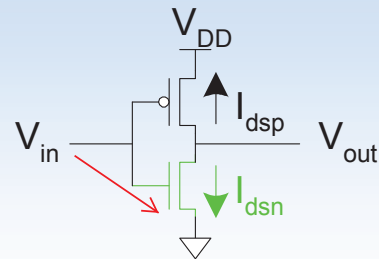
## nMOS Operation

| Cutoff      | Linear      | Saturated   |
|-------------|-------------|-------------|
| $V_{gsn} <$ | $V_{gsn} >$ | $V_{gsn} >$ |
|             | $V_{dsn} <$ | $V_{dsn} >$ |



# nMOS Operation

| Cutoff             | Linear                       | Saturated                    |
|--------------------|------------------------------|------------------------------|
| $V_{gsn} < V_{tn}$ | $V_{gsn} > V_{tn}$           | $V_{gsn} > V_{tn}$           |
|                    | $V_{dsn} < V_{gsn} - V_{tn}$ | $V_{dsn} > V_{gsn} - V_{tn}$ |

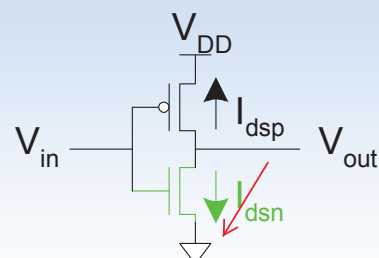


# nMOS Operation

| Cutoff             | Linear                       | Saturated                    |
|--------------------|------------------------------|------------------------------|
| $V_{gsn} < V_{tn}$ | $V_{gsn} > V_{tn}$           | $V_{gsn} > V_{tn}$           |
|                    | $V_{dsn} < V_{gsn} - V_{tn}$ | $V_{dsn} > V_{gsn} - V_{tn}$ |

$$V_{gsn} = V_{in}$$

$$V_{dsn} = V_{out}$$

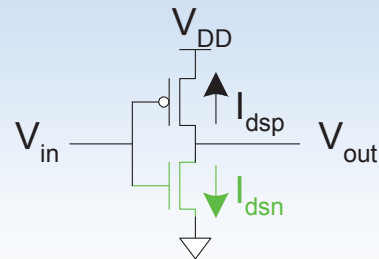


## nMOS Operation

| Cutoff                                  | Linear                                                                                                 | Saturated                                                                                              |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| $V_{gsn} < V_{tn}$<br>$V_{in} < V_{tn}$ | $V_{gsn} > V_{tn}$<br>$V_{in} > V_{tn}$<br>$V_{dsn} < V_{gsn} - V_{tn}$<br>$V_{out} < V_{in} - V_{tn}$ | $V_{gsn} > V_{tn}$<br>$V_{in} > V_{tn}$<br>$V_{dsn} > V_{gsn} - V_{tn}$<br>$V_{out} > V_{in} - V_{tn}$ |

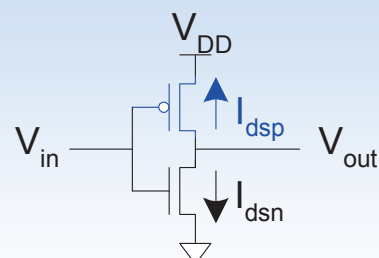
$$V_{gsn} = V_{in}$$

$$V_{dsn} = V_{out}$$



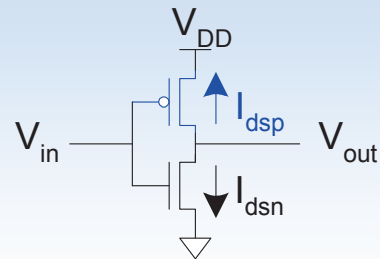
## pMOS Operation

| Cutoff      | Linear                     | Saturated                  |
|-------------|----------------------------|----------------------------|
| $V_{gsp} >$ | $V_{gsp} <$<br>$V_{dsp} >$ | $V_{gsp} <$<br>$V_{dsp} <$ |



## pMOS Operation

| Cutoff             | Linear                       | Saturated                    |
|--------------------|------------------------------|------------------------------|
| $V_{gsp} > V_{tp}$ | $V_{gsp} < V_{tp}$           | $V_{gsp} < V_{tp}$           |
|                    | $V_{dsp} > V_{gsp} - V_{tp}$ | $V_{dsp} < V_{gsp} - V_{tp}$ |



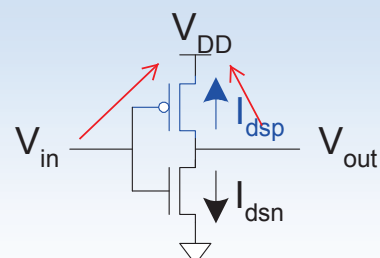
## pMOS Operation

| Cutoff             | Linear                       | Saturated                    |
|--------------------|------------------------------|------------------------------|
| $V_{gsp} > V_{tp}$ | $V_{gsp} < V_{tp}$           | $V_{gsp} < V_{tp}$           |
|                    | $V_{dsp} > V_{gsp} - V_{tp}$ | $V_{dsp} < V_{gsp} - V_{tp}$ |

$$V_{gsp} = V_{in} - V_{DD}$$

$$V_{tp} < 0$$

$$V_{dsp} = V_{out} - V_{DD}$$



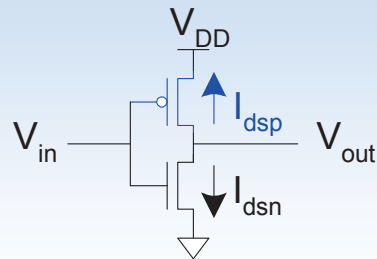
# pMOS Operation

| Cutoff                                           | Linear                                                                                                          | Saturated                                                                                                       |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| $V_{gsp} > V_{tp}$<br>$V_{in} > V_{DD} + V_{tp}$ | $V_{gsp} < V_{tp}$<br>$V_{in} < V_{DD} + V_{tp}$<br>$V_{dsp} > V_{gsp} - V_{tp}$<br>$V_{out} > V_{in} - V_{tp}$ | $V_{gsp} < V_{tp}$<br>$V_{in} < V_{DD} + V_{tp}$<br>$V_{dsp} < V_{gsp} - V_{tp}$<br>$V_{out} < V_{in} - V_{tp}$ |

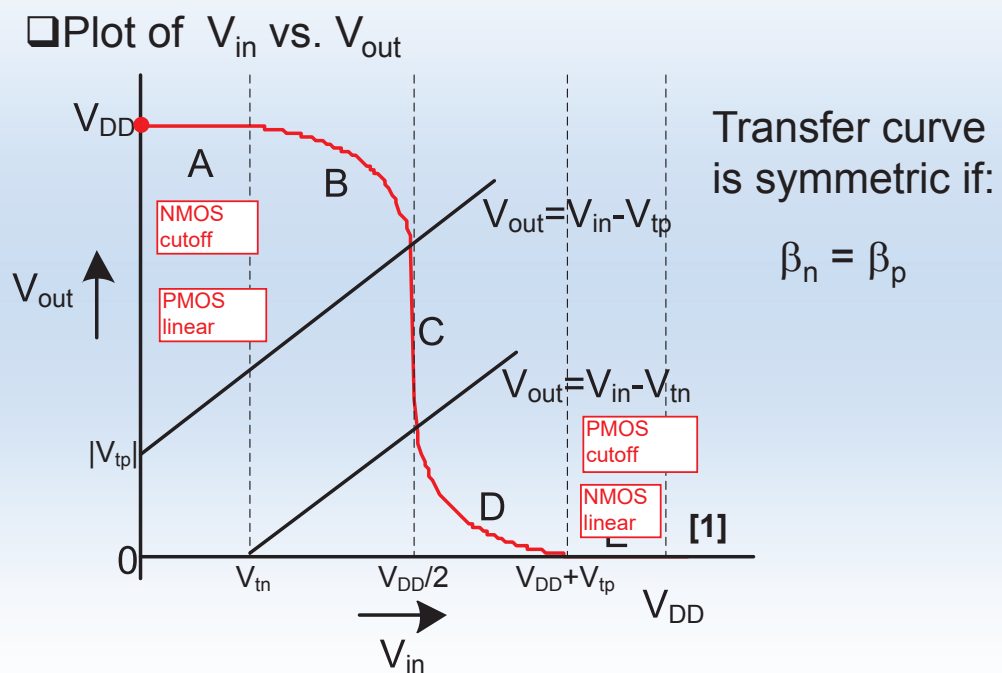
[1]

$$V_{\text{gsp}} = V_{\text{in}} - V_{\text{DD}} \quad V_{\text{tp}} < 0$$

$$V_{\text{dsp}} = V_{\text{out}} - V_{\text{DD}}$$



# DC Transfer Curve

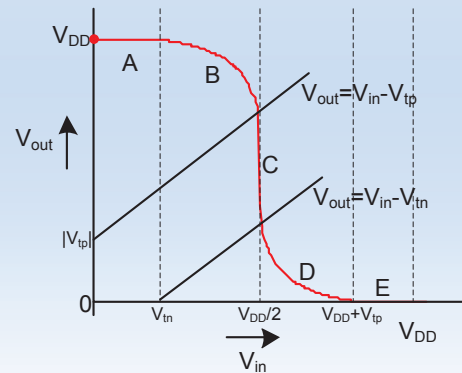




# Operating Regions

- Revisit transistor operating regions

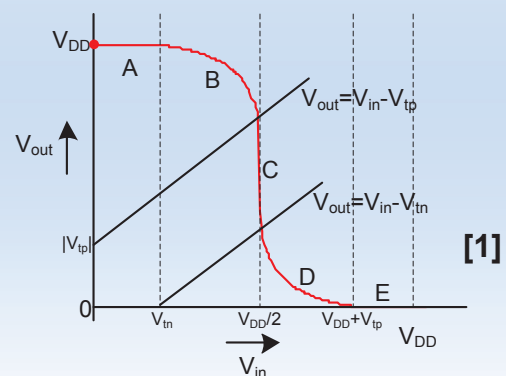
| Region | nMOS | pMOS |
|--------|------|------|
| A      |      |      |
| B      |      |      |
| C      |      |      |
| D      |      |      |
| E      |      |      |



# Operating Regions

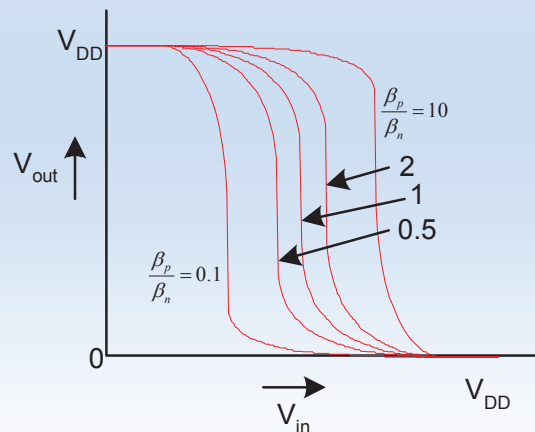
- Revisit transistor operating regions

| Region | nMOS       | pMOS       |
|--------|------------|------------|
| A      | Cutoff     | Linear     |
| B      | Saturation | Linear     |
| C      | Saturation | Saturation |
| D      | Linear     | Saturation |
| E      | Linear     | Cutoff     |



## Beta Ratio

- If  $\beta_p / \beta_n \neq 1$ , switching point will move from  $V_{DD}/2$



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## CMOS-Inverter: Delay Estimation

- Use RC delay models to estimate delay

$C$  = total capacitance on output node

Use *effective resistance*  $R$

So that  $t_{pd} = RC$

- Use equivalent circuits for MOS transistors

Ideal switch + capacitance and ON resistance

Unit nMOS has resistance  $R$ , capacitance  $C$

Unit pMOS has resistance  $2R$ , capacitance  $C$

- Capacitance proportional to width

- Resistance inversely proportional to width

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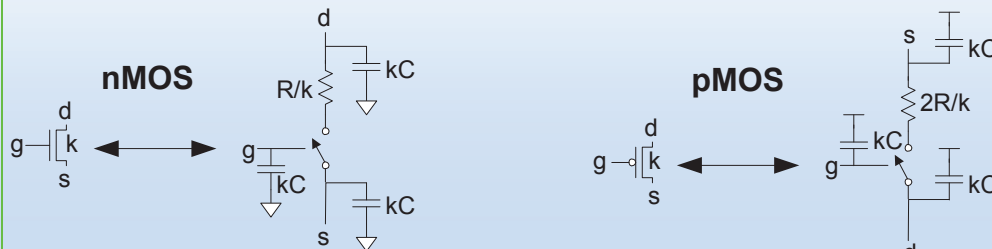
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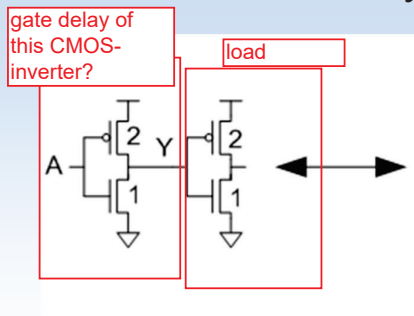
# CMOS-Inverter: Delay Estimation

NMOS: current flow of electrons

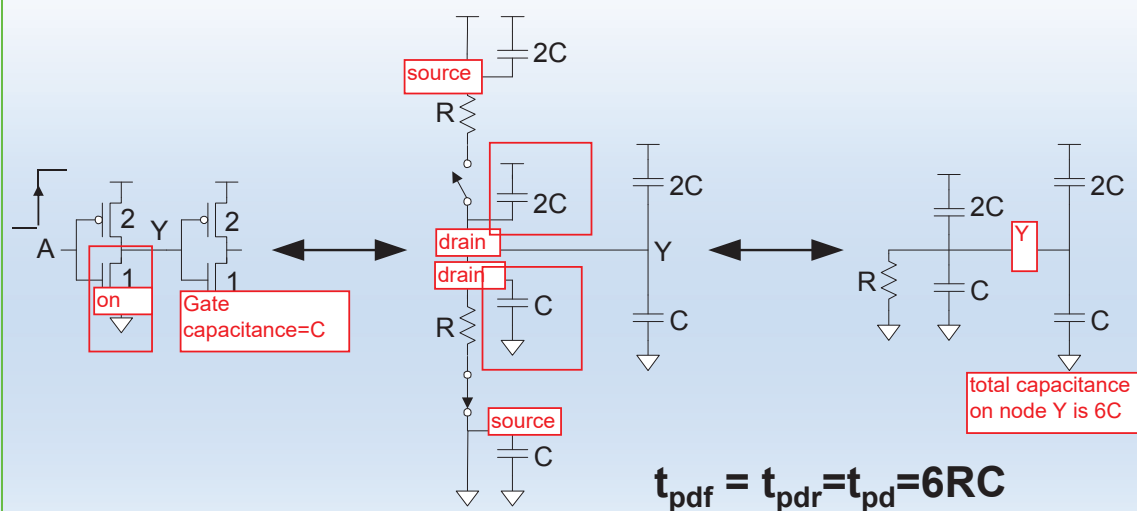
PMOS: current flow of holes



Estimate the delay of a fanout-of-1 inverter



# CMOS-Inverter: Delay Estimation



Capacitance  $C_g = C_s = C_d = C = 2 \text{ fF}/\mu\text{m}$  of gate width

Resistance  $R \approx 3 \text{ K}\Omega \cdot \mu\text{m}$  for 180nm CMOS-process

## Delay Definition

□  $t_{pdr}$ : *rising propagation delay*

– From input to rising output crossing  $V_{DD}/2$

□  $t_{pdf}$ : *falling propagation delay*

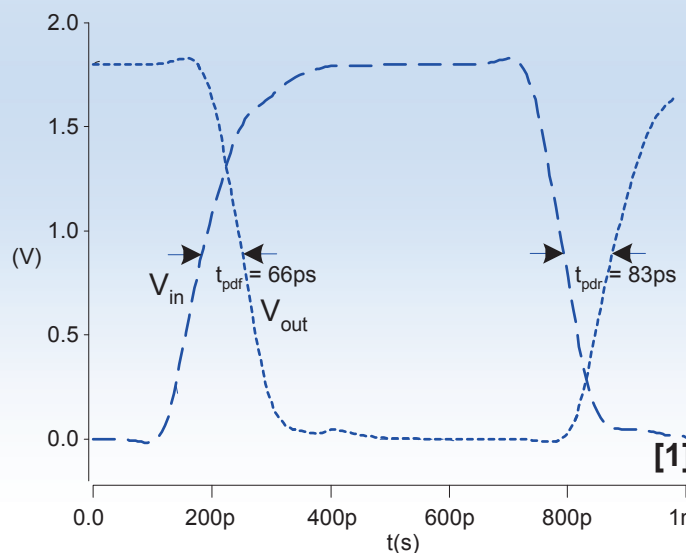
– From input to falling output crossing  $V_{DD}/2$

□  $t_{pd}$ : *average propagation delay*

$$t_{pd} = (t_{pdr} + t_{pdf})/2$$

## Simulated Inverter Delay

- Solving differential equations by hand is too hard
- SPICE simulator solves the equations numerically
  - Uses more accurate I-V models too!
- But simulations take time to write



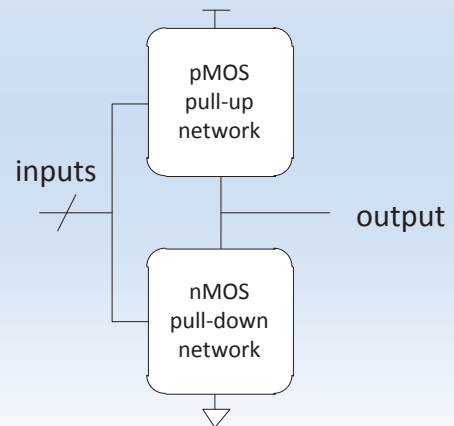
# Basic Gates in CMOS

## 1. Complementary CMOS Logic (Static CMOS Logic)

### □ Complementary CMOS logic gates (static CMOS gates)

nMOS *pull-down network*

pMOS *pull-up network*



### □ Complementary CMOS gates always produce 0 or 1

#### □ Ex: NAND gate

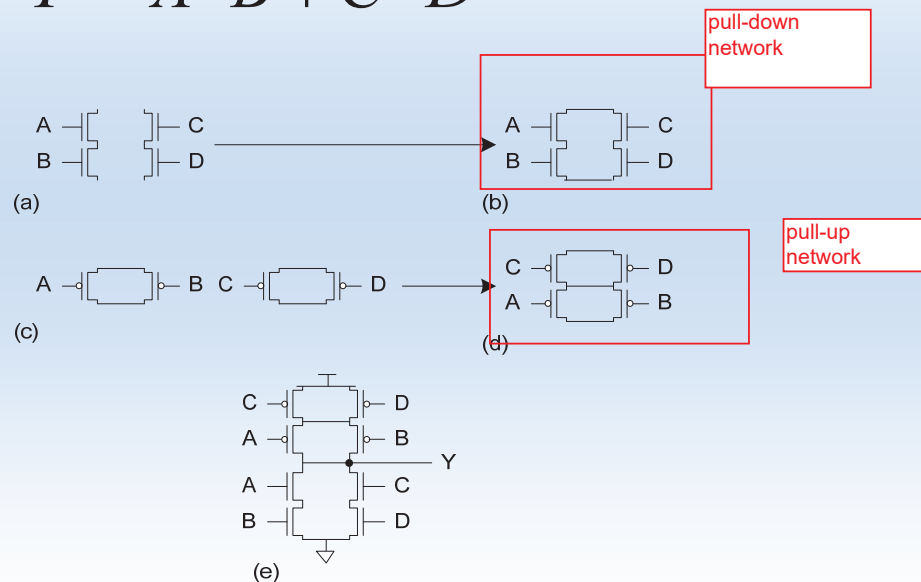
- Series nMOS:  $Y=0$  when both inputs are 1
- Thus  $Y=1$  when either input is 0
- Requires parallel pMOS

#### □ Rule of *Conduction Complements*

- Pull-up network is complement of pull-down
- parallel  $\rightarrow$  series, series  $\rightarrow$  parallel

□ Complementary CMOS logic gates can do any inverting function

□ Ex:  $Y = \overline{A \cdot B + C \cdot D}$



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## Example: O3AI

$$Y = \overline{(A + B + C) \cdot D}$$

Draw a transistor schematic in complementary CMOS logic!

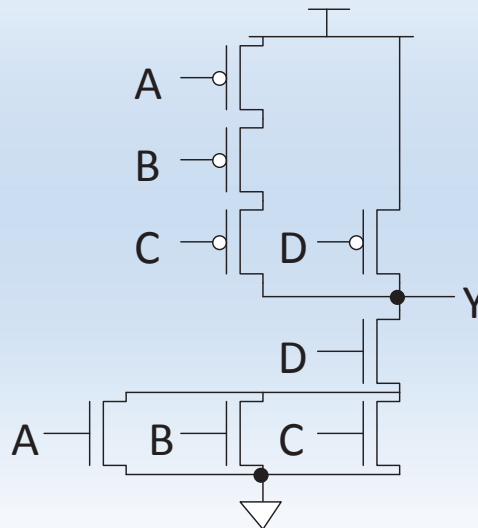
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## Example: O3AI

$$Y = \overline{(A + B + C) \cdot D}$$



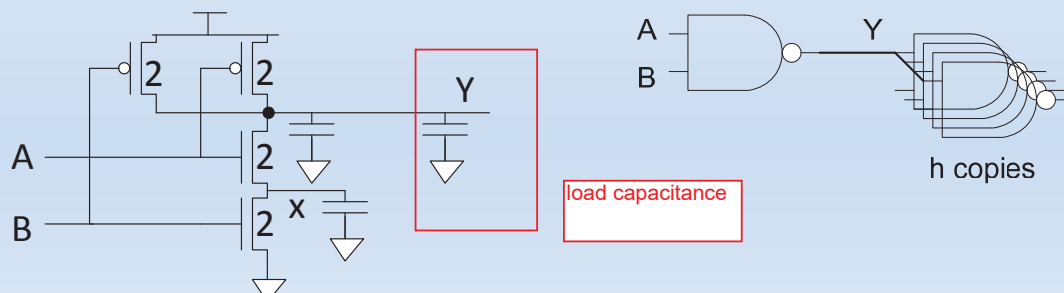
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## Example: 2-input NAND

- Estimate worst-case rising and falling delay of 2-input NAND driving  $h$  identical gates.



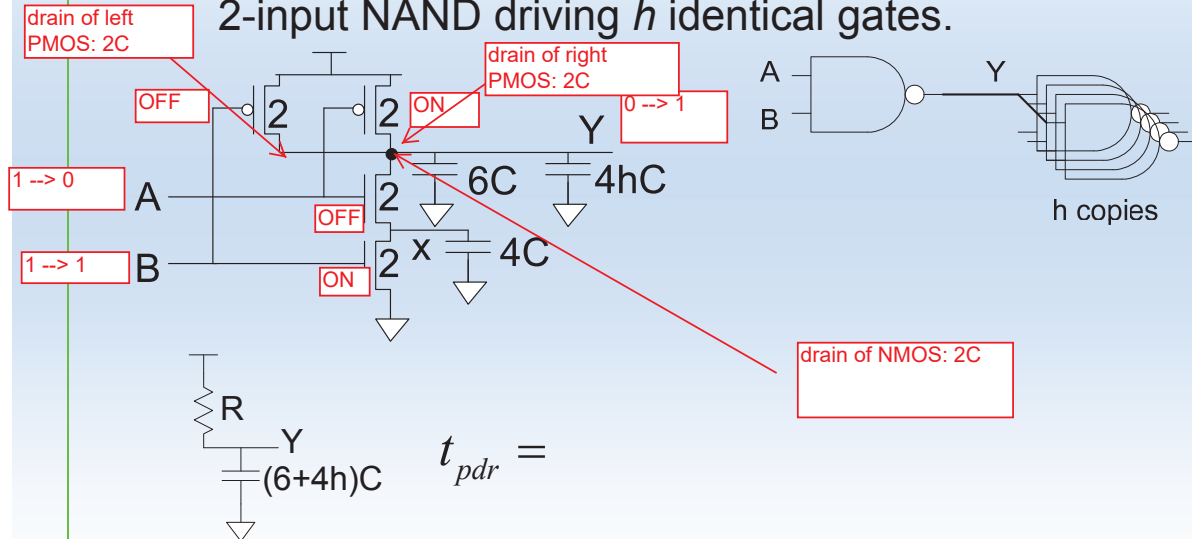
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## Example: 2-input NAND

- Estimate worst-case **rising** propagation delays of a 2-input NAND driving  $h$  identical gates.



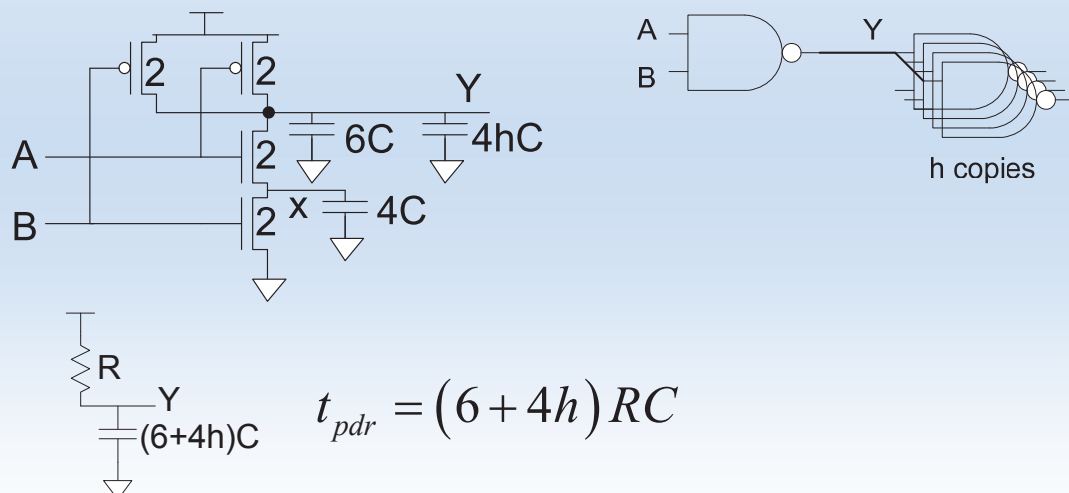
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## Example: 2-input NAND

- Estimate worst-case **rising** propagation delays of a 2-input NAND driving  $h$  identical gates.



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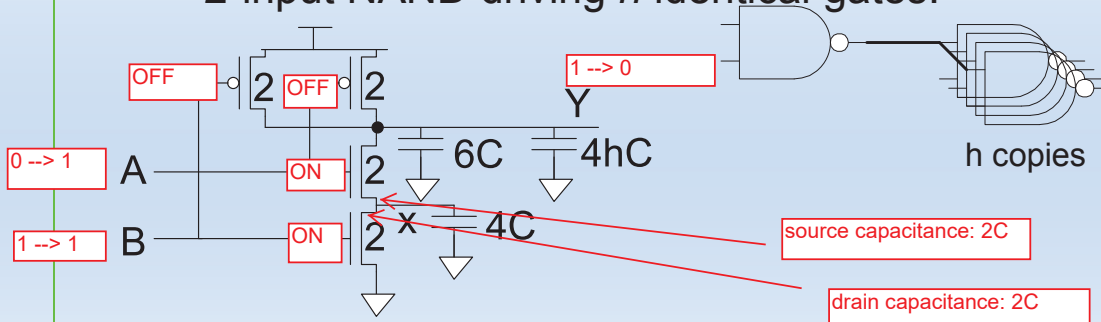
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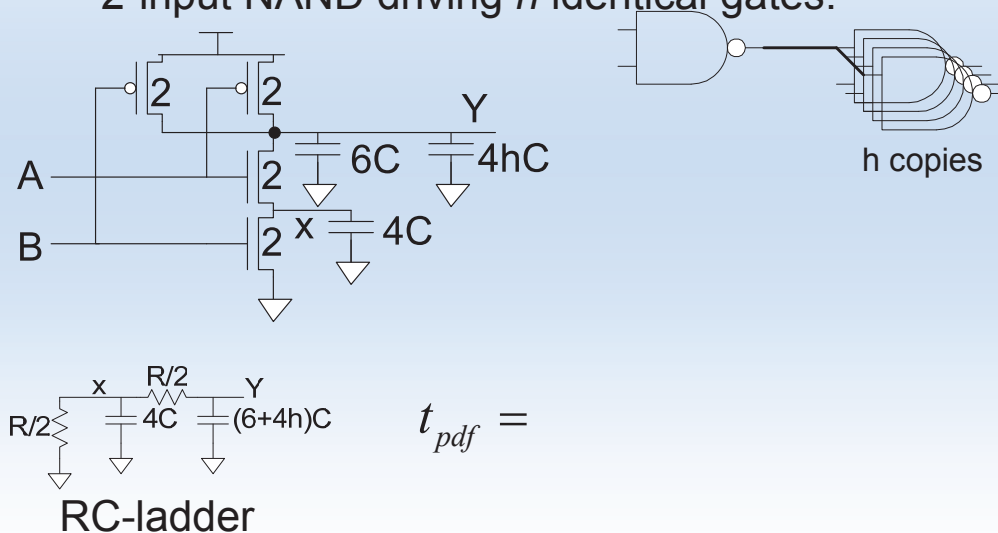
## Example: 2-input NAND

- Estimate worst-case **falling** propagation delays of a 2-input NAND driving  $h$  identical gates.



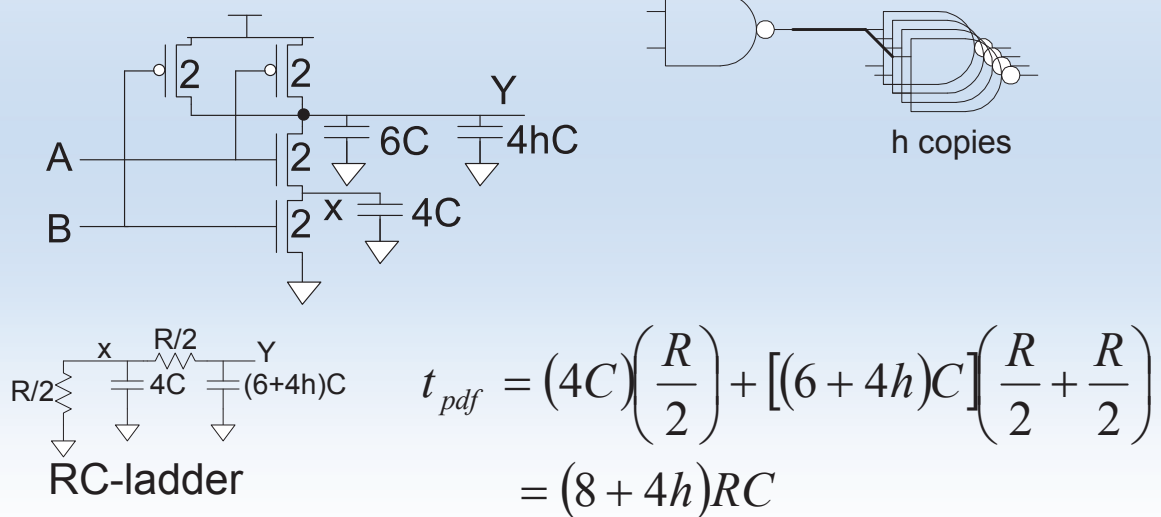
## Example: 2-input NAND

- Estimate worst-case **falling** propagation delays of a 2-input NAND driving  $h$  identical gates.



## Example: 2-input NAND

- Estimate worst-case **falling** propagation delays of a 2-input NAND driving  $h$  identical gates.



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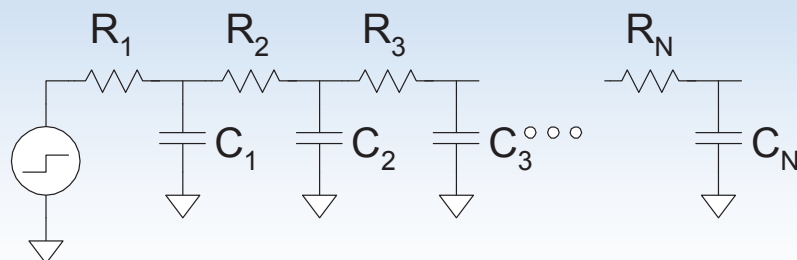
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## RC-Ladder: Elmore Delay

- Pullup or pulldown network modeled as *RC-ladder*
- Elmore delay of RC-ladder

$$t_{pd} \approx \sum_{\text{nodes } i} R_{i\text{-to-source}} C_i$$

$$= R_1 C_1 + (R_1 + R_2) C_2 + \dots + (R_1 + R_2 + \dots + R_N) C_N$$



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# Delay Components

- Delay has two parts
  - *Parasitic delay*
    - 6 or 8 RC
    - Independent of load
  - *Effort delay*
    - 4h RC
    - Proportional to load capacitance

## 2. Pass Transistor Logic

Pass Transistor Logic uses  
Transfer-Gates or Transmission-Gates

□ nMOS passing  $V_{DD}$

□  $V_g = V_{DD}$

– If  $V_s > V_{DD} - V_t$ ,  $V_{gs} < V_t$

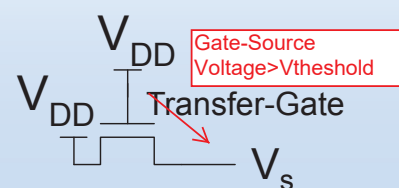
– Hence transistor would turn itself off

□ nMOS pass transistors pull no higher than  $V_{DD} - V_{tn}$

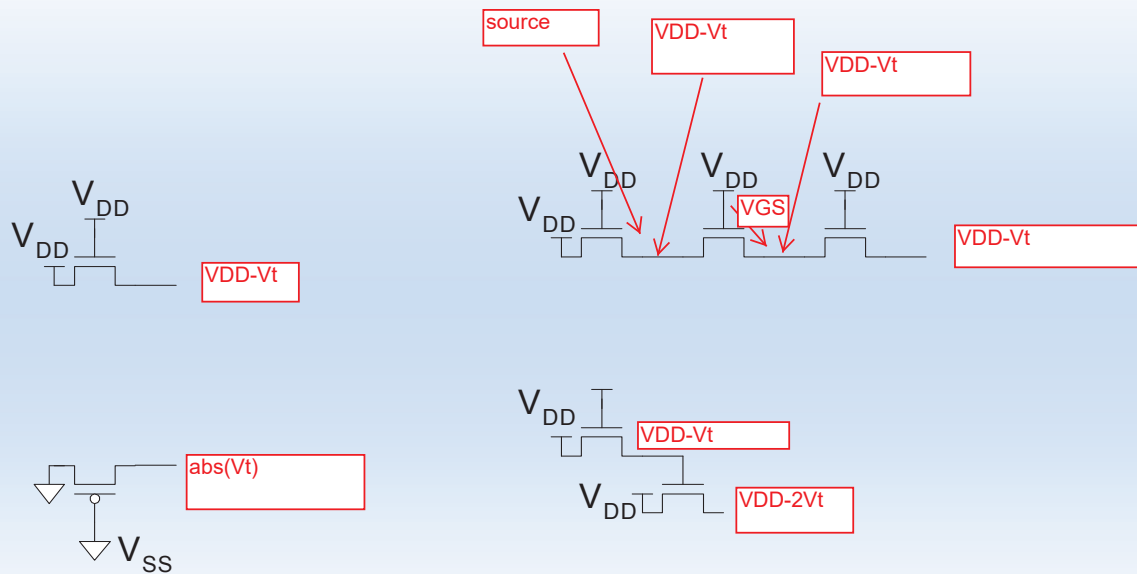
– Called a degraded “1”

□ pMOS pass transistors pull no lower than  $|V_{tp}|$

– Called a degraded “0”



# Circuit Design using Transfer-Gates

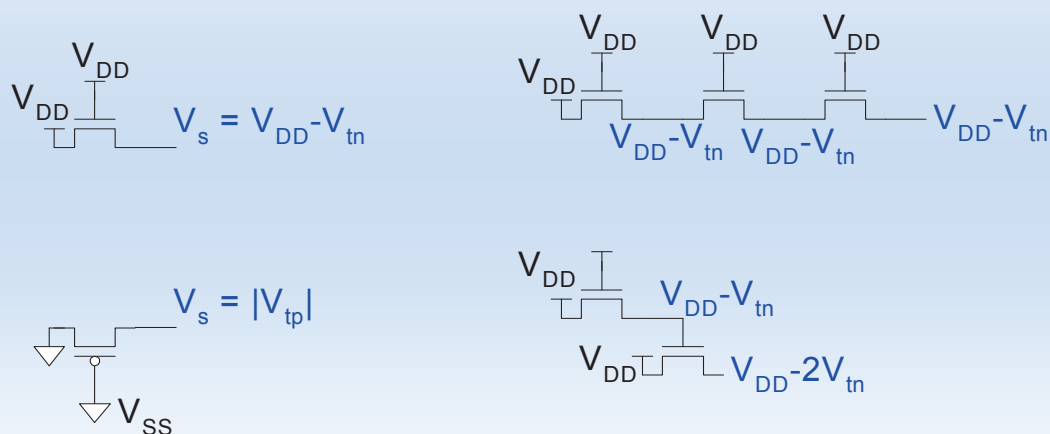


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# Circuit Design using Transfer-Gates



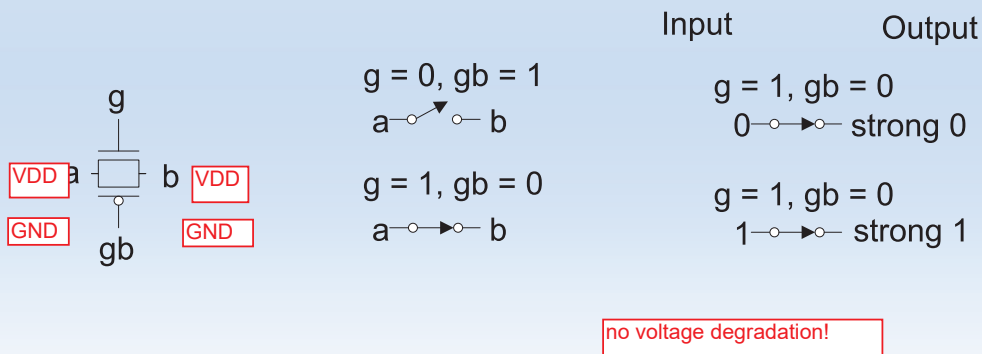
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# Transmission-Gate

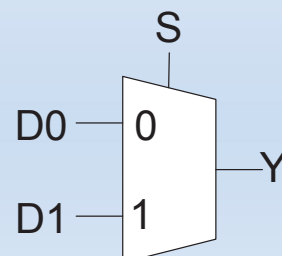
- *Transmission-gates* pass both 0 and 1 well
- *Transmission-gate* a parallel connection of nMOS and pMOS, both working as transfer-gates



# Multiplexer

- 2:1 *multiplexer* chooses between two inputs

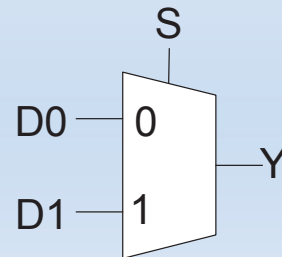
| S | D1 | D0 | Y |
|---|----|----|---|
| 0 | X  | 0  |   |
| 0 | X  | 1  |   |
| 1 | 0  | X  |   |
| 1 | 1  | X  |   |



# Multiplexer

- 2:1 multiplexer chooses between two inputs

| S | D1 | D0 | Y |
|---|----|----|---|
| 0 | X  | 0  | 0 |
| 0 | X  | 1  | 1 |
| 1 | 0  | X  | 0 |
| 1 | 1  | X  | 1 |

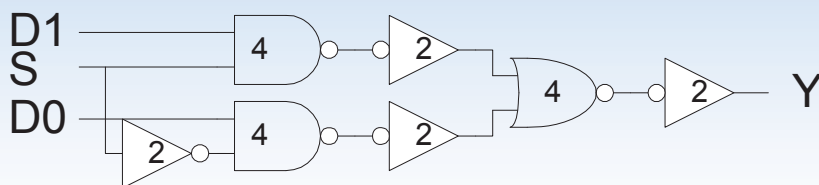
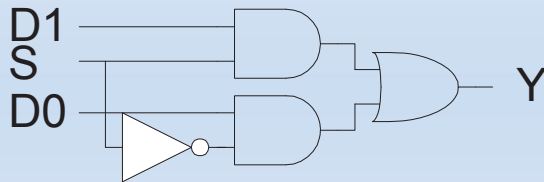


# Gate-Level Mux Design

- $Y = SD_1 + \overline{S}D_0$  (too many transistors)
- How many transistors are needed?

## Gate-Level Mux Design

- $Y = SD_1 + \bar{S}D_0$  (too many transistors)
- How many transistors are needed? 20



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## Transmission Gate Mux

- Nonrestoring mux uses two transmission gates

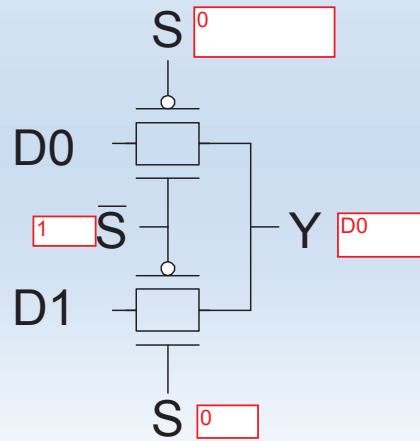
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## Transmission-Gate Mux

- Nonrestoring mux uses two transmission gates
  - Only 4 transistors



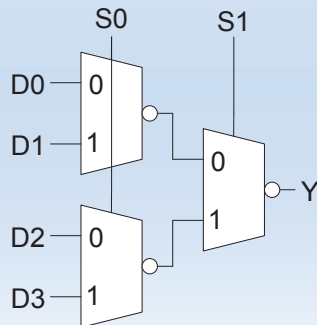
## 4:1 Multiplexer

- 4:1 mux chooses one of 4 inputs using two selects



## 4:1 Multiplexer

- 4:1 mux chooses one of 4 inputs using two selects
  - Two levels of 2:1 muxes



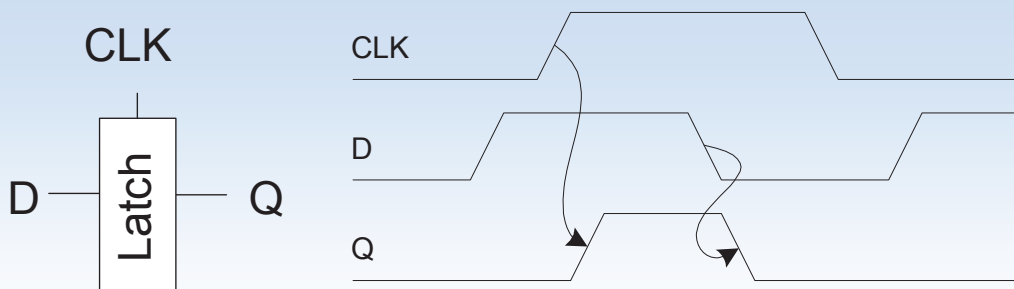
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## Transparent Latch

- When CLK = 1, latch is *transparent*
  - D flows through to Q like a buffer
- When CLK = 0, the latch is *opaque*
  - Q holds its old value independent of D
- Also called a *level-sensitive latch*



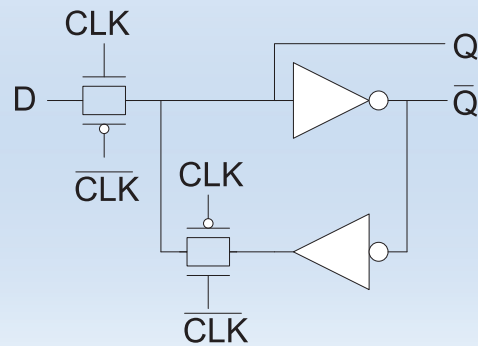
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# Latch Design

- Multiplexer chooses D or old Q

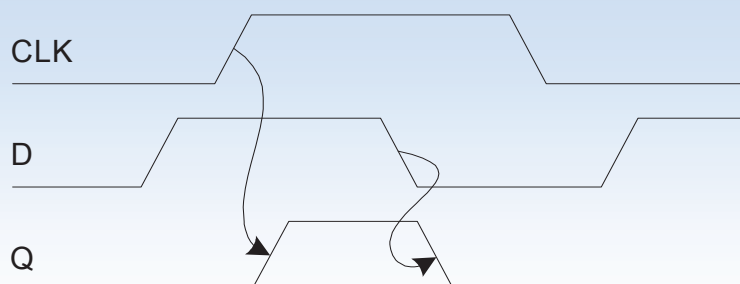
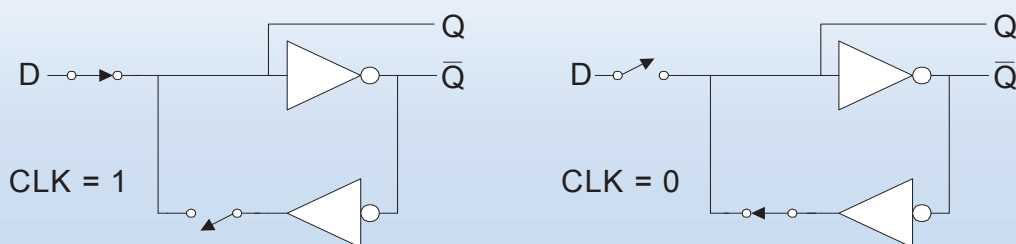


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# Latch Operation



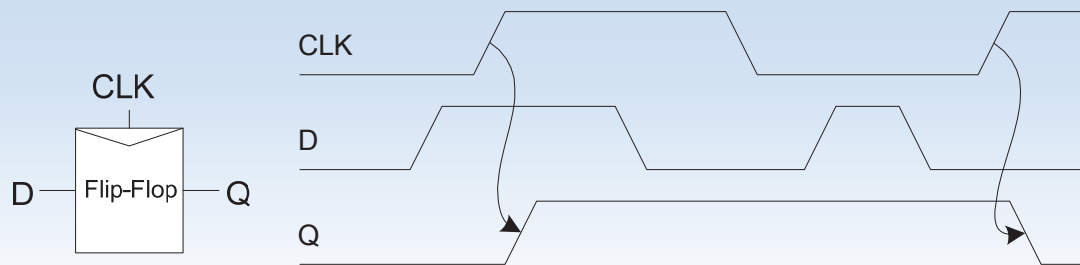
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# D-Flip-Flop

- When CLK rises, D is copied to Q
- At all other times, Q holds its value
- a.k.a. *positive edge-triggered flip-flop, master-slave flip-flop*



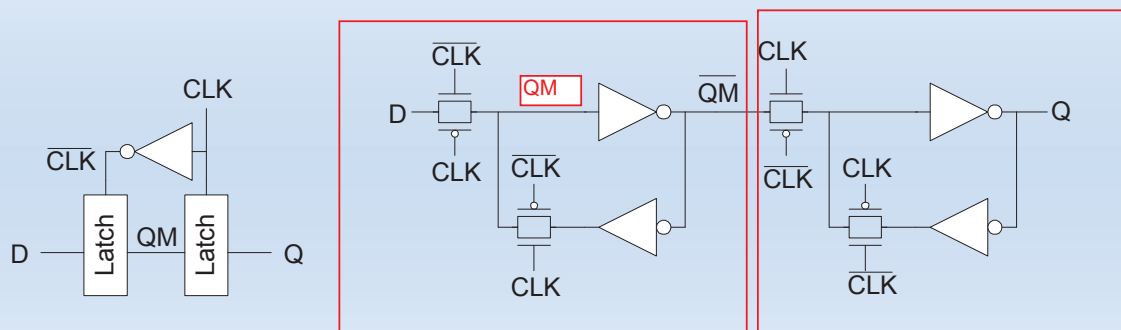
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# D-Flip-Flop Design

- Built from master and slave latches



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# D-Flip-Flop Operation

